Sharp logarithmic coupling estimates for theta equality frames in a signed Weil-form realization

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Abstract

We determine the size of the finite-frame Schur correction associated with the odd theta derivatives in a weighted realization of Weil's quadratic form. For the first m pole-killing odd derivatives, the exact correction satisfies

$$\|E_{m,\beta}\| \asymp_m \beta\big(\log(1/\beta)\big)^{4m+5/2} \qquad (\beta \downarrow 0).$$

Both inequalities are unconditional, with explicit constants and parameter regimes. The upper bound follows from a constructive cancellation of the source charge on each half-line, a square-root range argument, and an integrated estimate of the full theta tail. Odd box tests give the matching lower bound. We also obtain a pointwise upper bound for the low-energy spectral mass of the weighted gamma operator. The prime-shift operator and the polar terms are retained in the exact Schur reduction. The estimate rules out an $O_m(\beta)$ bound for this correction; it does not determine the sign of the remaining complementary form. An exact change of metric also identifies how eliminating the equality space before gamma normalization cancels the explicit correction and moves its logarithmic growth into the metric.

1 The estimate and its setting

We use

$$\xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \qquad \widehat{f}(t) = \int_{\mathbb{R}} f(x)e^{itx}\,dx.$$

The Fourier convention has Plancherel factor $1/(2\pi)$. Inner products are linear in the first entry. The actual even theta kernel is

$$\Phi(x) = \sum_{n\geq 1}(4\pi^2 n^4 e^{9|x|/2} - 6\pi n^2 e^{5|x|/2})e^{-\pi n^2 e^{2|x|}}, \qquad \widehat{\Phi}(t) = \xi(1/2+it). \tag{1}$$

It is smooth and positive. Its smooth even extension and Fourier normalization are classical; see Romik [2, (1.8)–(1.11), printed p. 2]. Put

$$c(x) = \cosh(x/2), \qquad w(x)^2 = \Phi(x)/c(x), \qquad H = L^2(\mathbb{R},\,dx).$$

Let $\psi = \Gamma'/\Gamma$, and define the complete gamma symbols

$$\begin{aligned} m_\Gamma(t) &= \Re\psi(1/4+it/2) - \log\pi, \\ c_\Gamma &= m_\Gamma(0) = -\gamma_E - \pi/2 - 3\log 2 - \log\pi, \\ b(t) &= m_\Gamma(t) - c_\Gamma = \sum_{j\geq 0}\frac{(t/2)^2}{(j+1/4)\{(j+1/4)^2+(t/2)^2\}}. \end{aligned} \tag{2}$$

Define G through its maximal closed nonnegative form

$$g(u) = \frac{1}{2\pi}\int_{\mathbb{R}} b(t)|\widehat{wu}(t)|^2\,dt, \qquad \mathcal{V} = \{u \in H : g(u) < \infty\}. \tag{3}$$

The form domain, injectivity and relevant operator domains are proved below. No critical-line location of a zero is assumed.

For $r \geq 1$ set

$$d_r = 2r+1, \qquad \kappa_r = 4r + 5/2, \qquad h_r = \partial_x(\partial_x^2 - 1/4)^r\Phi, \qquad \widetilde{u}_r = h_r/w. \tag{4}$$

These are linearly independent odd vectors in $D(G)$. Fix $m \geq 1$, and choose any base-L^2 orthonormal basis $u_1, \dots, u_m$ of their first-m span. Write $U : \mathbb{C}^m \to H_{\mathrm{odd}}$ for the associated isometry, and put

$$\begin{aligned} F_\beta &= (G+\beta)^{-1/2}, & B_\beta &= (G+\beta)^{1/2}U, \\ H_0 &= U^*GU, & H_\beta &= H_0 + \beta I, \\ W_\beta &= B_\beta H_\beta^{-1/2}, & N_\beta &= I - W_\beta W_\beta^*. \end{aligned} \tag{5}$$

The finite matrix H_0 is positive definite. Thus W_β is an isometry, and N_β is an orthogonal projection. The finite-rank correction of interest is

$$E_{m,\beta} = \beta N_\beta F_\beta U U^* F_\beta N_\beta \quad \text{on } H_{\mathrm{odd}}. \tag{6}$$

Theorem 1.1 (Sharp fixed-frame coupling)**.** *For every fixed $m \geq 1$ there are explicitly defined positive constants a_m, A_m, ℓ_m such that*

$$a_m\beta\big(\log(1/\beta)\big)^{\kappa_m} \leq \|E_{m,\beta}\| \leq A_m\beta\big(\log(1/\beta)\big)^{\kappa_m} \quad (0 < \beta \leq e^{-\ell_m}). \tag{7}$$

One may take the constants in (32)–(33)*. In the same regime,*

$$\|U^*\mathbf{1}_{(0,t]}(G)U\| \leq 2A_m t\big(\log(1/t)\big)^{\kappa_m} \quad (0 < t \leq e^{-\ell_m}). \tag{8}$$

The exact norm identity is

$$\|E_{m,\beta}\| = \beta\|U^*(G+\beta)^{-1}U - H_\beta^{-1}\|. \tag{9}$$

In particular, the first vector gives the exponent $13/2$. The constants and the small-parameter regime depend on m. Equation (8) is an upper bound for a discontinuous spectral projection; a matching pointwise lower bound is not asserted.

The estimate concerns a specific full-symbol resolvent and its exact finite-dimensional correction. The Weil and Li frameworks themselves have substantial antecedents. Bombieri–Lagarias [1, Theorem 1 and (2.4); Section 4] give the explicit-formula relation underlying Li's criterion. Suzuki's norm formula [3, Theorem 1.1, (1.8)] is equivalent to RH, and its orthonormal-basis argument [3, Proposition 3.2] assumes RH. Suzuki's localized Weil operators [5, Theorem 1.1; Sections 2.4–2.5] provide the local gamma and prime formulas used here. Our weighted whole-line G is instead defined directly by the closed form (3). The proof below quantifies its actual theta-jet resolvent, then identifies the correction in the full signed Weil form. No inner-function or conditional model-space property is imported. Invertible-block Schur elimination is standard; compare Dusson–Sigal–Stamm [7, Theorem 1.2, (1.10)–(1.11)]. The quantitative step here is the actual theta-weighted resolvent estimate. The zero-charge primitive is a Fourier antiderivative construction, used with separate half-line compensation to obtain the stated rate.

2 The complete weighted gamma form

Lemma 2.1 (Theta bounds). *With $C_- = 4\pi^2 - 6\pi > 0$ and $C_\Phi = 8\pi^2$, for every real x,*

$$C_- e^{9|x|/2-\pi e^{2|x|}} \le \Phi(x) \le C_\Phi e^{9|x|/2-\pi e^{2|x|}}, \tag{10}$$

$$C_- e^{4|x|-\pi e^{2|x|}} \le w(x)^2 \le 2C_\Phi e^{4|x|-\pi e^{2|x|}}. \tag{11}$$

For every fixed $j \ge 0$, a finite constant C_j gives

$$|\Phi^{(j)}(x)| \le C_j e^{(9/2+2j)|x|-\pi e^{2|x|}}.$$

Proof. For $x \ge 0$, every theta summand is positive. Its $n = 1$ term gives the lower bound. For the upper bound discard the negative terms and factor out $e^{9x/2-\pi e^{2x}}$. The remaining sum is $4\pi^2 \sum n^4 e^{-\pi(n^2-1)e^{2x}}$. For $n \ge 2$, $n^4 \le 16\,6^{n-2}$ and $n^2 - 1 \ge n + 1$. Since $q = e^{-\pi} < 1/20$, this tail is at most $16q^3/(1-6q) < 1$. The total sum is less than two. Divide by $c(x)$, using $e^{x/2}/2 \le c(x) \le e^{x/2}$. Evenness supplies the other half-line.

Each fixed derivative of a summand is a polynomial of degree at most j in $\pi n^2 e^{2x}$, times the original exponential. Factoring the first exponential leaves a summable Gaussian n-polynomial majorant. This proves the derivative bound and justifies termwise differentiation locally. In particular every fixed derivative is Schwartz. □

Lemma 2.2 (Closed form, jump formula and domains). *The form* (3) *is densely defined and closed, and its associated self-adjoint operator G is nonnegative and injective. For $f = wu$,*

$$g(u) = \frac{1}{2} \int_{\mathbb{R}} \frac{e^{-|a|/2}}{1 - e^{-2|a|}} \int_{\mathbb{R}} |f(x+a) - f(x)|^2 \, dx \, da. \tag{12}$$

Compact smooth functions are a form core. If $u \in \mathcal{V}$ and the physical Fourier multiplier $b(D_x)(wu)$ belongs to L^2, then

$$u \in D(G), \qquad Gu = w\, b(D_x)(wu). \tag{$*$}$$

Moreover, at every $t \ne 0$,

$$b(t) \ge \frac{16t^2}{1+4t^2}, \qquad b(t)^{-1} \le \frac{1}{4} + \frac{1}{16t^2}. \tag{13}$$

Proof. The digamma series gives (2); its first nonnegative summand gives (13). Also

$$b(t) = \int_{\mathbb{R}} \mathfrak{h}(a)(1 - \cos ta)\, da, \qquad \mathfrak{h}(a) = \frac{e^{-|a|/2}}{1 - e^{-2|a|}}.$$

Indeed expand $\mathfrak{h} = \sum_{j \ge 0} e^{-(2j+1/2)|a|}$, integrate each nonnegative term, and obtain (2). Tonelli and Plancherel prove (12). The series also shows $b(t) = O(\log(2 + |t|))$: split it at $j + 1/4 = |t|/2$, bounding the first part by a harmonic sum and the second by a cubic tail.

Multiplication by the bounded w maps H continuously into L^2. The map $u \mapsto b(D_x)^{1/2}(wu)$ is closed: convergence of u in H implies convergence of wu, and the physical multiplier is closed. This proves closedness of (3). Compact smooth functions give density. If $g(u) = 0$, then $\widehat{wu}$ is supported on $\{0\}$, because $b(t) > 0$ off zero. An L^2 function with that support vanishes. Since $w > 0$, $u = 0$.

Here are details of the core assertion. Let χ_R be smooth, equal to one on $[-R, R]$, supported on $[-2R, 2R]$, and with $|\chi_R'| \le C/R$, for $R \ge 1$. In the jump formula for $(1 - \chi_R)f$, split its difference into a cutoff times the difference of f, and f times a cutoff difference. Dominated convergence handles the first term. The second tends to zero and is dominated by $C'\|f\|_2^2 \mathfrak{h}(a) \min(a^2, 1)$, an

integrable function. Thus $\chi_R u \to u$ in the form norm. For a fixed R, mollify the compact physical function $f_R = w\chi_R u$. Fourier dominated convergence gives convergence of the mollification in the b-form norm as well as in L^2. Divide the mollified functions by w; on the fixed enlarged compact support, $1/w$ is smooth and bounded. This gives compact smooth approximants to $\chi_R u$ in the required form norm. Finally, if $b(D_x)(wu) \in L^2$, Plancherel gives the polarized form as $\langle wb(D_x)(wu), v\rangle$ for every $v \in \mathcal{V}$. The form representation theorem proves $(*)$. □

3 Charge cancellation and the resolvent upper bound

Lemma 3.1 (Constructive square-root range approximation)**.** *Suppose $u \in H$, $a = u/w$ is odd, and $|a(x)| \le Ce^{\kappa|x|}$, where $C, \kappa > 0$. Complex-valued odd sources are allowed. Put*

$$R_\kappa = \max\left(1, \frac{1}{2}\log\frac{2(\kappa+1)}{\pi}\right).$$

For $R \ge R_\kappa$, define

$$\begin{aligned} &\alpha_R = \int_0^R a(x)\,dx, && b_R = \mathbf{1}_{[R+1,R+2]} - \mathbf{1}_{[-R-2,-R-1]}, \\ &a_R = a\mathbf{1}_{[-R,R]} - \alpha_R b_R, && u_R = wa_R, \qquad P_R(x) = \int_0^x a_R(t)\,dt. \end{aligned} \tag{14}$$

Both half-line integrals of a_R vanish. The vector u_R belongs to $\operatorname{Ran} G^{1/2}$*, and*

$$\|G^{-1/2}u_R\|^2 \le K_A e^{2\kappa R}, \tag{15}$$

$$\|u - u_R\|^2 \le K_T e^{(2\kappa+2)R - \pi e^{2R}}, \tag{16}$$

$$K_A = C^2\left(\frac{1}{4\kappa} + \frac{3}{4\kappa^2} + \frac{1}{16\kappa^3}\right),$$

$$K_T = 8C_\Phi C^2\left(\frac{1}{\pi} + \frac{1}{\kappa^2}\right).$$

Proof. Oddness cancels the negative half-line charge once the positive one is canceled. The primitive P_R is even and supported in $[-R-2, R+2]$. On $[0,R]$, $|P_R(x)| \le Ce^{\kappa x}/\kappa$; on $[R, R+1]$ it is α_R; on $[R+1, R+2]$ it decreases linearly to zero. The supports of the two pieces of a_R are disjoint. Therefore

$$\|a_R\|^2 \le C^2(\kappa^{-1} + 2\kappa^{-2})e^{2\kappa R},$$
$$\|P_R\|^2 \le C^2(\kappa^{-3} + 4\kappa^{-2})e^{2\kappa R}.$$

The derivative $P_R' = a_R$ holds in distributions and almost everywhere. Hence $\widehat{a}_R = -it\widehat{P}_R$, and (13) gives

$$\|b(D_x)^{-1/2}a_R\|^2 \le \tfrac{1}{4}\|a_R\|^2 + \tfrac{1}{16}\|P_R\|^2 \le K_A e^{2\kappa R}.$$

For every $v \in \mathcal{V}$, Fourier Cauchy–Schwarz yields

$$|\langle u_R, v\rangle| = |\langle a_R, wv\rangle| \le \|b(D_x)^{-1/2}a_R\|\,\|G^{1/2}v\|.$$

Injectivity makes $\operatorname{Ran} G^{1/2}$ dense. The bounded functional on this range has a Riesz representative q_R of norm at most $\|b(D_x)^{-1/2}a_R\|$. Thus $\langle u_R, v\rangle = \langle q_R, G^{1/2}v\rangle$ on the entire form domain. The

adjoint-domain characterization of the self-adjoint $G^{1/2}$ gives $q_R \in D(G^{1/2})$ and $G^{1/2}q_R = u_R$. This proves (15) without taking the formal inverse of a product of unbounded operators.

Write $Y = e^{2R}$. By (11), the omitted source tail has weighted square norm at most

$$4C_\Phi C^2 \int_R^\infty e^{(2\kappa+4)x-\pi e^{2x}}\,dx = 2C_\Phi C^2 \int_Y^\infty y^{\kappa+1}e^{-\pi y}\,dy \le \frac{4C_\Phi C^2}{\pi} Y^{\kappa+1}e^{-\pi Y}.$$

Here $\pi Y \ge 2(\kappa+1)$, so the logarithmic derivative of the last integrand is at most $-\pi/2$. The weighted square norm of the outer charge rectangles is at most

$$\frac{4C_\Phi C^2}{\kappa^2} e^{(2\kappa+4)R+4-\pi e^2 Y} \le \frac{4C_\Phi C^2}{\kappa^2} Y^{\kappa+1}e^{-\pi Y}.$$

For the final inequality use $Ye^4e^{-\pi(e^2-1)Y} \le 1$ for $Y \ge 1$; $\log Y \le Y$, $e^2 > 7$, and $\pi > 3$ suffice. The square triangle inequality proves (16). The half-line cancellation keeps the primitive's large plateau in two outer strips of bounded length, instead of across the central interval. This is why its norm has no additional factor R. □

Proposition 3.2 (Explicit upper rate)**.** *Under the hypotheses of Lemma 3.1, define*

$$\ell_\kappa = \max(\pi e^2/2, \kappa+1), \qquad K_U(C,\kappa) = 2(K_A+K_T)(2/\pi)^\kappa. \tag{17}$$

For $0 < \beta \le e^{-\ell_\kappa}$,

$$\mathcal{R}_u(\beta) := \langle u, (G+\beta)^{-1}u\rangle \le K_U(C,\kappa)\big(\log(1/\beta)\big)^\kappa.$$

Also, for $0 < t \le e^{-\ell_\kappa}$,

$$\langle \mathbf{1}_{(0,t]}(G)u, u\rangle \le 2K_U(C,\kappa)t\big(\log(1/t)\big)^\kappa.$$

Proof. The resolvent square-root triangle inequality gives

$$\mathcal{R}_u(\beta) \le 2K_A e^{2\kappa R} + 2\beta^{-1}K_T e^{(2\kappa+2)R-\pi e^{2R}}.$$

Set $L = \log(1/\beta)$ and $e^{2R} = 2L/\pi$. The stated regime is exactly $R \ge R_\kappa$. The error term is $2K_T\beta(2L/\pi)^{\kappa+1}$. Since $\beta(2L/\pi) < 1$, (17) gives the first bound. For the second apply $\mathbf{1}_{(0,t]}(s) \le 2t/(s+t)$ in the spectral theorem. No assumption on the type of the spectral measure is used. □

4 Actual theta jets and matching lower bounds

The following finite definitions supply constants for every jet. For $\alpha = 9/2, 5/2$, let

$$P_{\alpha,0}(q) = 1, \qquad P_{\alpha,k+1}(q) = (\alpha-2q)P_{\alpha,k}(q) + 2qP'_{\alpha,k}(q),$$
$$R_{\alpha,r}(q) = \sum_{k=0}^{r}\binom{r}{k}(-1/4)^{r-k}P_{\alpha,2k+1}(q), \qquad B_{\alpha,r} = \sum_{j=0}^{d_r}|[q^j]R_{\alpha,r}(q)|\,\pi^{j-d_r}. \tag{18}$$

For a positive integer p, set

$$M = p+1, \qquad \rho_p = e^{1-\pi(2M+1)}, \qquad \mathcal{S}_p = \sum_{n=1}^{M-1} n^p e^{-\pi(n^2-1)} + \frac{M^p e^{-\pi(M^2-1)}}{1-\rho_p}. \tag{19}$$

Finally put

$$C_{a,r} = \frac{\pi^{d_r}}{C_-}\left(4\pi^2 B_{9/2,r}\mathcal{S}_{4+2d_r} + 6\pi B_{5/2,r}\mathcal{S}_{2+2d_r}\right). \tag{20}$$

Lemma 4.1 (Whole-lattice jet bounds)**.** *The constants* (18)–(20) *give*

$$\left|\frac{h_r(x)}{\Phi(x)}\right| \le C_{a,r}e^{2d_r|x|}, \qquad \left|\frac{\widetilde{u}_r(x)}{w(x)}\right| \le C_{a,r}e^{\kappa_r|x|}. \tag{21}$$

Every $\widetilde{u}_r$ *belongs to* $D(G)$*. Moreover, as* $x \to +\infty$*,*

$$\frac{c(x)h_r(x)}{\Phi(x)} = -\frac{(2\pi)^{d_r}}{2}e^{\kappa_r x}(1+e^{-x})(1+O_r(e^{-2x})). \tag{22}$$

Consequently the vectors $\widetilde{u}_1, \dots, \widetilde{u}_m$ *are independent for every* m*.*

Proof. The recurrence in (18) is differentiation of $e^{\alpha x - \pi n^2 e^{2x}}$. The polynomial $R_{\alpha,r}$ has degree d_r and leading coefficient -2^{d_r}. For $q \ge \pi$, $|R_{\alpha,r}(q)| \le B_{\alpha,r}q^{d_r}$. The nth differentiated theta term is

$$4\pi^2 n^4 e^{9x/2-q_n}R_{9/2,r}(q_n) - 6\pi n^2 e^{5x/2-q_n}R_{5/2,r}(q_n), \qquad q_n = \pi n^2 e^{2x}.$$

For $n \ge M = p+1$, the ratio of successive terms of $n^p e^{-\pi(n^2-1)}$ is at most $e^{p/n-\pi(2n+1)} \le \rho_p < 1$. Thus $\mathcal{S}_p$ bounds the complete Gaussian sum. Divide the derivative series by the positive lower bound (10), and use $e^{-\pi(n^2-1)e^{2x}} \le e^{-\pi(n^2-1)}$ for $x \ge 0$. This proves the first inequality in (21). Multiplication by $c(x) \le e^{|x|/2}$ proves the second. Parity supplies $x < 0$.

The lower weight bound and the derivative bounds show $\widetilde{u}_r = h_r/w \in L^2$. The physical function h_r is Schwartz, and $b(t) = O(\log(2+|t|))$, so $b(D_x)h_r \in L^2$. Lemma 2.2 puts $\widetilde{u}_r$ in $D(G)$.

In the $n = 1$ term, take the leading coefficients of the finite polynomials and divide by the leading term of Φ. The relative polynomial error is $O_r(e^{-2x})$. For $n \ge 2$, the complete relative remainder is bounded by a constant times

$$\sum_{n\ge 2} n^{4+2d_r}e^{-\pi(n^2-1)e^{2x}} \le \mathcal{S}_{4+2d_r}e^{-3\pi(e^{2x}-1)}.$$

It is smaller than the stated polynomial error. Multiplication by $c(x) = e^{x/2}(1+e^{-x})/2$ proves (22) with the full lattice retained. A nontrivial finite linear combination has a unique largest κ_r; its leading tail cannot vanish. This proves independence. □

To specify the lower regime, define

$$\begin{aligned}
S_{\alpha,r} &= \sum_{j<d_r} |[q^j]R_{\alpha,r}(q)|, \\
Q_r &= \max(1, 2^{1-d_r}S_{9/2,r}, 2^{1-d_r}S_{5/2,r}), \\
R_{0,r} &= \max(1, \tfrac{1}{2}\log(Q_r/\pi)), & c_{a,r} &= 2^{d_r-4}\pi^{d_r}, \\
R_{*,r} &= R_{0,r} + \kappa_r^{-1}\log\frac{2(C_{a,r}+c_{a,r})}{c_{a,r}}, \\
\overline{C}_g &= 218/25, & \ell_{L,r} &= 2\pi e^{2R_{*,r}}, \\
K_{L,r} &= \frac{c_{a,r}^2}{\kappa_r^2(4\overline{C}_g+1)(2\pi)^{\kappa_r}}.
\end{aligned} \tag{23}$$

Proposition 4.2 (Explicit matching lower rate)**.** *For every* $r \ge 1$ *and* $0 < \beta \le e^{-\ell_{L,r}}$*,*

$$\mathcal{R}_{\widetilde{u}_r}(\beta) \ge K_{L,r}\big(\log(1/\beta)\big)^{\kappa_r}.$$

Proof. For $q \ge Q_r$, $q \ge 1$ and the sum of all lower-degree terms is at most $S_{\alpha,r} q^{d_r-1}$. Therefore

$$-3 \cdot 2^{d_r-1} q^{d_r} \le R_{\alpha,r}(q) \le -2^{d_r-1} q^{d_r}.$$

For $x \ge R_{0,r}$, this holds for every q_n. In each differentiated theta summand, the ratio of the possible positive contribution from the second term to the negative leading bound from the first is at most

$$\frac{9}{2\pi n^2} e^{-2x} < \tfrac{1}{2}.$$

Thus all derivative summands are negative. Retaining half of the first negative bound and then the $n = 1$ term yields

$$h_r(x) \le -2^{d_r} \pi^{d_r+2} e^{(9/2+2d_r)x - \pi e^{2x}}.$$

Divide this negative numerator by the *upper* bound on Φ, and use $c(x) \ge e^{x/2}/2$. With $a_r = c h_r/\Phi = \widetilde{u}_r/w$, we obtain

$$a_r(x) \le -c_{a,r} e^{\kappa_r x} \quad (x \ge R_{0,r}). \tag{24}$$

Let $H_R(x) = \operatorname{sgn}(x) \mathbf{1}_{[-R,R]}(x)$ and $v_R = H_R/w$. For finite R, $v_R \in L^2$. The physical box H_R has supremum one and total variation four. Hence

$$\int |H_R(x+a) - H_R(x)|^2 \, dx \le 8|a|.$$

The full jump formula gives

$$g(v_R) \le 4C_g, \qquad C_g = \int_{\mathbb{R}} \mathfrak{h}(a)|a| \, da = 2 \sum_{j \ge 0} (2j + 1/2)^{-2} \le 8 + 2(4/25 + 1/5) = \overline{C}_g.$$

This also proves $v_R \in \mathcal{V}$, including its jumps. The upper central bound (21) and the negative tail (24) imply

$$\langle \widetilde{u}_r, v_R \rangle = 2 \int_0^R a_r(x) \, dx \le -\frac{c_{a,r}}{\kappa_r} e^{\kappa_r R} \quad (R \ge R_{*,r}).$$

Indeed the central integral contributes at most

$$\frac{2C_{a,r}}{\kappa_r} e^{\kappa_r R_{0,r}},$$

while the tail contributes at most

$$-\frac{2c_{a,r}}{\kappa_r} \big(e^{\kappa_r R} - e^{\kappa_r R_{0,r}} \big).$$

The definition of $R_{*,r}$ gives the displayed inequality.

The lower weight bound, and monotonicity of $e^{-4x+\pi e^{2x}}$ on $x \ge 0$, give

$$\|v_R\|^2 \le \frac{2R}{C_-} e^{-4R + \pi e^{2R}}.$$

Set $L = \log(1/\beta)$ and $e^{2R} = L/(2\pi)$. Then $R \ge R_{*,r} \ge 1$ in the stated regime, and

$$\beta \|v_R\|^2 \le \frac{2R}{C_-} e^{-4R - L/2} \le \frac{2e^{-4}}{C_-} < 1.$$

The resolvent variational inequality on the form domain now gives

$$\mathcal{R}_{\widetilde{u}_r}(\beta) \ge \frac{|\langle \widetilde{u}_r, v_R \rangle|^2}{g(v_R) + \beta \|v_R\|^2} \ge \frac{c_{a,r}^2}{\kappa_r^2 (4\overline{C}_g + 1)} \left(\frac{L}{2\pi} \right)^{\kappa_r}.$$

It follows directly by Cauchy–Schwarz applied to $(G+\beta)^{-1/2} \widetilde{u}_r$ and $(G+\beta)^{1/2} v_R$. This proves the proposition with no truncation of the gamma kernel. □

5 The exact signed Weil coupling

This section identifies the correction (6) in the complete Weil form, including the two polar moments. It is not needed for the scalar resolvent estimates.

Define the bounded self-adjoint full prime-shift operator by

$$(Ju)(x) = \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}} w(x)\{w(x-\log n)u(x-\log n) + w(x+\log n)u(x+\log n)\}. \tag{25}$$

For $q \in H$, let $(q \otimes q)u = \langle u, q\rangle q$. Put

$$\begin{aligned} &p_c = w\cosh(x/2), && p_s = w\sinh(x/2), \\ &\mathcal{P} = 2(p_c \otimes p_c - p_s \otimes p_s), && D = J - c_\Gamma M_w^2, \qquad D_W = D - \mathcal{P}. \end{aligned} \tag{26}$$

Both pole vectors belong to H.

Lemma 5.1 (Full operator and unconditional equality)**.** *The series* (25) *converges in operator norm, and D_W is bounded and self-adjoint. On compact smooth physical tests $f = wu$, the Weil form is exactly*

$$Q_W(f) = g(u) - \langle D_W u, u\rangle. \tag{27}$$

This formula supplies a closed lower-bounded weighted extension. Every vector in (4) *satisfies, unconditionally,*

$$\mathcal{P}\widetilde{u}_r = 0, \qquad (G - D)\widetilde{u}_r = (G - D_W)\widetilde{u}_r = 0. \tag{28}$$

Proof. For $a = \log n$, set $A = e^{2|x|}$, $B = e^{2|x-a|}$. By (11),

$$w(x)w(x-a) \leq 2C_\Phi AB e^{-\pi(A+B)/2} \leq C_\Phi e^{-\pi(A+B)/8} \leq C_\Phi e^{-\pi n/4}.$$

Here $A, B \geq 1$, $\log(2AB) \leq A + B < 3\pi(A+B)/8$, and $AB \geq n^2$, so $A + B \geq 2n$. The paired shift has operator norm at most $2C_\Phi e^{-\pi n/4}$. Since $\Lambda(n) \leq \log n$, $\sum_{n\geq 2} 2C_\Phi (\log n) n^{-1/2} e^{-\pi n/4} < \infty$. The two shifts are adjoints of one another, proving the assertion. The remaining terms in (26) are bounded self-adjoint operators.

For completeness, specify the local explicit formula used here. Let $\mathcal{Z}$ be the distinct zeros, with multiplicities m_γ, of $X(z) = \xi(1/2 + iz)$. They satisfy $|\Im\gamma| < 1/2$ without RH. For $b_0 \in C_c^\infty(\mathbb{R})$, Weil's explicit formula gives the local distribution

$$\langle k, b_0\rangle = \sum_{\gamma\in\mathcal{Z}} m_\gamma \widehat{b}_0(\gamma), \qquad k = 2\cosh(\,\cdot\,/2) + K_\Gamma - \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}}(\delta_{\log n} + \delta_{-\log n}), \tag{29}$$

where K_Γ is the tempered Fourier multiplier $m_\Gamma(D_x)$, with its finite-part kernel at zero. The zero sum is absolutely convergent for these tests. This is the unconditional compact-test formula; see Suzuki [4, Theorem 1.1(2), (1.3)] and [5, Sections 2.4–2.5; (3.1), printed p. 13]. Only K_Γ, not the whole k, is assigned a global tempered Fourier multiplier. One can obtain its linear zero formula directly from the cited unconditional expansion

$$\Psi(t) = \sum_{\gamma\in\mathcal{Z}} m_\gamma \frac{1 - \cos(\gamma t)}{\gamma^2}, \qquad k = \Psi''.$$

The series converges absolutely and uniformly on compact real intervals, by the strip bound and $\sum m_\gamma(1+|\gamma|)^{-2} < \infty$. There is no zero at 0, since $X(0) = \int \Phi > 0$. Pair it with b_0'', integrate twice

by parts, and use the symmetry of the full zero multiset under $\gamma \mapsto -\gamma$. This gives exactly (29), without a half-set factor. The contact normalization can also be specified directly:

$$\begin{aligned}\langle K_\Gamma, b_0\rangle = & -(\log(4\pi)+\gamma_E)b_0(0)\\ & -\int_0^\infty \{b_0(x)+b_0(-x)-2e^{-x/2}b_0(0)\}\frac{e^{-x/2}}{1-e^{-2x}}\,dx.\end{aligned}$$

The subtraction makes the integral convergent at zero. The digamma integral confirms its Fourier symbol m_Γ; the constants agree because

$$2\int_0^\infty \frac{e^{-x}-e^{-2x}}{1-e^{-2x}}\,dx = 2\log 2 = \log 4.$$

Suzuki's 2026 xi normalization omits the factor $1/2$. That changes neither the zeros, their multiplicities nor this explicit formula; (1) retains the normalization used here.

Apply (29) to the correlation $b_0(t) = \int f(x)\overline{f(x-t)}\,dx$. Equivalently, for a compact smooth f,

$$Q_W(f) = \sum_{\gamma\in\mathcal{Z}} m_\gamma \widehat{f}(\gamma)\overline{\widehat{f}(\bar\gamma)}.$$

This sum is absolutely convergent by compact-test Fourier decay, the strip bound and zero counting. Multiplicities are retained. The polar contribution is

$$2\left|\int f(x)\cosh(x/2)\,dx\right|^2 - 2\left|\int f(x)\sinh(x/2)\,dx\right|^2.$$

The gamma contribution is $(2\pi)^{-1}\int m_\Gamma(t)|\widehat{f}(t)|^2\,dt$; the prime contribution is $-\langle Ju,u\rangle$. Splitting $m_\Gamma = b + c_\Gamma$ proves (27). Bounded perturbation of the closed form g gives its stated weighted extension. The zero formula itself is only being invoked on the compact tests.

Here is an extension of the local formula sufficient for equality. For a fixed real x, test with a compact smooth cutoff of $t \mapsto h_r(x-t)$. For each derivative of order at most two, the cutoff remainder tends to zero in $L^1(e^{|t|/2}\,dt)$, by Lemma 2.1. Two integrations by parts and the bound on $|\Im\gamma|$ give

$$|\widehat{b_{x,R}}(\gamma) - \widehat{h_r(x-\cdot)}(\gamma)| \le \epsilon_{x,R}(1+|\gamma|)^{-2}, \qquad \epsilon_{x,R}\to 0.$$

Zero counting gives $\sum m_\gamma(1+|\gamma|)^{-2} < \infty$. The limiting transform is zero at every γ, since

$$\widehat{h}_r(z) = (-iz)(-z^2-1/4)^r X(z).$$

Evenness of X also makes it zero at $-\gamma$. Thus the zero side tends absolutely to zero. On the physical side the polar term is integrable against h_r; the gamma cutoff converges in Schwartz seminorms; and the entire prime series converges locally absolutely, with all fixed derivatives, by its theta decay and $\Lambda(n) \le \log n$. Hence $k * h_r = 0$ locally on the whole real line. No rearrangement into an unproved global Fourier transform of k has been made.

Integration by parts gives

$$\int h_r(x)e^{\pm x/2}\,dx = 0: \quad (\partial_x^2 - 1/4)^r e^{\pm x/2} = 0 \quad (r\ge 1).$$

There are no boundary terms, by the full theta decay. Both pole moments therefore vanish, and (29) gives

$$K_\Gamma h_r = \sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}}\{h_r(\cdot-\log n)+h_r(\cdot+\log n)\}.$$

Multiply by w. Lemma 4.1 already puts $\widetilde{u}_r$ in $D(G)$; the right side is the norm-convergent operator $J\widetilde{u}_r$. Using $K_\Gamma = b(D_x) + c_\Gamma$ proves (28). □

For this full signed operator, let

$$T_\beta = I - F_\beta D_W F_\beta \quad \text{on } H_{\text{odd}}. \tag{30}$$

All its factors are bounded for each fixed $\beta > 0$. Reflection preserves the gamma form, the prime operator and the pole operator, so the odd restriction is valid.

Proposition 5.2 (Exact finite elimination)**.** *The positive finite equality block is*

$$W_\beta^* T_\beta W_\beta = \beta H_\beta^{-1}.$$

Its Schur complement on $N_\beta H_{\text{odd}}$ is exactly

$$\mathcal{C}_{m,\beta} = N_\beta T_\beta N_\beta - E_{m,\beta}. \tag{31}$$

In particular their negative indices agree: $n_-(T_\beta) = n_-(\mathcal{C}_{m,\beta})$. Identity (9) *holds, and its matrix difference is nonnegative. The same correction results if D_W is replaced by the gamma-prime operator D.*

Proof. Equality (28) gives

$$T_\beta B_\beta = \beta F_\beta U, \qquad B_\beta^* F_\beta U = I.$$

Thus $W_\beta^* T_\beta W_\beta = \beta H_\beta^{-1} > 0$. In the orthogonal decomposition $H_{\text{odd}} = \operatorname{Ran} W_\beta \oplus N_\beta H_{\text{odd}}$, complete the square with the inverse $\beta^{-1} H_\beta$ of this finite block. The off-diagonal block is $\beta N_\beta F_\beta U H_\beta^{-1/2}$. Its product with the inverse pivot is exactly (6); the two factors $H_\beta^{\pm 1/2}$ cancel. The completion is a bounded invertible triangular congruence for each fixed β, proving the negative-index assertion.

The nonzero eigenvalues of $N_\beta F_\beta U U^* F_\beta N_\beta$ equal those of

$$U^* F_\beta N_\beta F_\beta U = U^* (G + \beta)^{-1} U - H_\beta^{-1} \geq 0.$$

This proves (9). The calculation of this matrix needs only the gamma form and its frame, rather than equality with a signed operator. Equality is needed to identify it as the Schur correction of T_β. Finally $\mathcal{P} U = 0$, so both signed choices satisfy the same identity $T_\beta B_\beta = \beta F_\beta U$. The complementary block still contains every prime and polar term. □

6 Proof of the fixed-frame rate and explicit constants

Write $u_j = \sum_{r=1}^m c_{jr} \widetilde{u}_r$, and define

$$\begin{aligned} C_j &= \sum_{r=1}^m |c_{jr}| C_{a,r}, & A_m &= \sum_{j=1}^m K_U(C_j, \kappa_m), \\ q_m &= \|\widetilde{u}_m\|^2, & \mu_m &= \lambda_{\min}(H_0) > 0, \\ a_m &= \frac{K_{L,m}}{2q_m}. \end{aligned} \tag{32}$$

One explicit regime is

$$\ell_m = \max\left(\ell_{\kappa_m}, \ell_{L,m}, \left(\frac{2q_m}{K_{L,m} \mu_m} \right)^{1/\kappa_m} \right). \tag{33}$$

All definitions in these formulas are specified in (18)–(23) and (17).

Proof of Theorem 1.1. Injectivity of G and independence of the finite frame imply positive definiteness of H_0. Since $\kappa_r \le \kappa_m$,

$$|u_j(x)/w(x)| \le C_j e^{\kappa_m |x|}.$$

Proposition 3.2, positivity of the resolvent matrix and its trace yield

$$\|U^*(G+\beta)^{-1}U\| \le A_m\big(\log(1/\beta)\big)^{\kappa_m} \quad (\log(1/\beta) \ge \ell_{\kappa_m}).$$

The nonnegative matrix difference in (9) is bounded above by this matrix. This proves the upper estimate in (7). The same trace and spectral inequality prove (8).

The normalized vector $\widetilde{u}_m/\sqrt{q_m}$ is in the frame span. Proposition 4.2 gives

$$\|U^*(G+\beta)^{-1}U\| \ge \frac{K_{L,m}}{q_m}\big(\log(1/\beta)\big)^{\kappa_m}.$$

Also $\|H_\beta^{-1}\| \le \mu_m^{-1}$. The reverse norm inequality, and the last term of (33), therefore give

$$\|U^*(G+\beta)^{-1}U - H_\beta^{-1}\| \ge \frac{K_{L,m}}{2q_m}\big(\log(1/\beta)\big)^{\kappa_m}.$$

Multiply by β and use (9). Every bound holds on the stated full parameter interval. □

Remark 6.1 (Evaluation of the frame constants). The constants involve finite polynomials, finite sums, and convergent integrals and matrices. They do not require a numerical positivity assertion. For a definite choice of frame, let V have columns $\widetilde{u}_1, \dots, \widetilde{u}_m$, set $M_0 = V^*V$, and take $U = VM_0^{-1/2}$. Then $c_{jr} = (M_0^{-1/2})_{rj}$ and

$$H_0 = M_0^{-1/2} M_1 M_0^{-1/2}, \qquad M_1 = V^*GV.$$

The entries of M_0 are ordinary theta integrals. Those of M_1 are the polarized full gamma integrals $(2\pi)^{-1}\int b(t)\widehat{h}_r(t)\overline{\widehat{h}_s(t)}\,dt$. Their convergence was proved above. If an expression without a smallest-eigenvalue symbol is desired, use the explicit lower bound

$$\mu_m \ge \delta_m := \frac{\det H_0}{(\operatorname{tr} H_0)^{m-1}} > 0$$

and replace μ_m by δ_m in (33). The resulting larger threshold is valid for every m, including $m = 1$. These are effective fixed-frame definitions, with no asserted uniform conditioning as m grows.

Remark 6.2 (The first jet). The first polynomials are

$$R_{9/2,1} = 90 - 183q + 78q^2 - 8q^3, \qquad R_{5/2,1} = 15 - 75q + 54q^2 - 8q^3.$$

One may use the general constant (20), so no separate numerical bound is required. The common formula gives $\kappa_1 = 13/2$, and all first-jet constants and its parameter regime follow from the displayed finite definitions. There is also a shorter lower-tail estimate for this jet. For $q \ge 21$,

$$\begin{aligned} R_{9/2,1}(q) - \frac{3}{2q}R_{5/2,1}(q) + 4q^3\left(1 - \frac{3}{2q}\right) \\ = -4q^3 + 84q^2 - 264q + \frac{405}{2} - \frac{45}{2q} < 0. \end{aligned}$$

Since $\pi e^2 > 21$, every theta summand at $x \ge 1$ has derivative ratio at most $-4(\pi n^2 e^{2x})^3$. All original summands are positive, so summing proves

$$\frac{\Phi'''(x) - \Phi'(x)/4}{\Phi(x)} \le -4\pi^3 e^{6x}, \qquad \frac{c(x)(\Phi'''(x) - \Phi'(x)/4)}{\Phi(x)} \le -2\pi^3 e^{13x/2} \quad (x \ge 1).$$

Thus, for the first jet, the lower construction may instead use $R_{0,1} = 1$ and $c_{a,1} = 2\pi^3$, leaving (20) unchanged and recomputing $R_{*,1}, \ell_{L,1}, K_{L,1}$ by (23).

7 Normalization after equality-space elimination

There is an exact way to place the large correction in context. It eliminates the same equality space before gamma normalization. The resulting signed form still contains the full arithmetic operator, but the finite coupling term has disappeared.

Proposition 7.1 (Exact change of metric)**.** *Let $P_0 = UU^*$, $N_0 = I - P_0$ and $Y = N_0 H_{\mathrm{odd}}$. Let $G_\perp$ be the operator of the restricted gamma form on Y, and define*

$$C = N_0 G U, \qquad A_\perp = G_\perp - N_0 D_W N_0.$$

All occurrences of $N_0 D_W N_0$ below are restricted to Y. For $\beta > 0$ put

$$\begin{aligned} M_\beta &= G_\perp + \beta - C H_\beta^{-1} C^* \geq \beta, \\ J_\beta v &= v - U H_\beta^{-1} C^* v, \\ \mathcal{O}_\beta &= (G+\beta)^{1/2} J_\beta M_\beta^{-1/2}. \end{aligned} \tag{34}$$

The map $\mathcal{O}_\beta : Y \to N_\beta H_{\mathrm{odd}}$ is unitary, and the exact complement obeys

$$\mathcal{O}_\beta^* \mathcal{C}_{m,\beta} \mathcal{O}_\beta = M_\beta^{-1/2} (A_\perp + \beta) M_\beta^{-1/2}. \tag{35}$$

The right side is understood by its bounded form. With

$$\widehat{T}_\beta = (G_\perp + \beta)^{-1/2} (A_\perp + \beta)(G_\perp + \beta)^{-1/2}, \qquad V_\beta = (G_\perp + \beta)^{1/2} M_\beta^{-1/2},$$

both operators are bounded, V_β is invertible, and the right side of (35) *is $V_\beta^* \widehat{T}_\beta V_\beta$. Moreover*

$$\|V_\beta\|^2 - 1 = \left\| H_\beta^{1/2} \{ U^* (G+\beta)^{-1} U - H_\beta^{-1} \} H_\beta^{1/2} \right\| \asymp_m \left(\log(1/\beta) \right)^{\kappa_m}. \tag{36}$$

Proof. The finite projection preserves the gamma form domain because its columns lie in $D(G)$. The restricted form on Y is closed and densely defined. Its off-diagonal gamma block is the bounded map C, so the gamma block matrix is

$$G + \beta = \begin{pmatrix} H_\beta & C^* \\ C & G_\perp + \beta \end{pmatrix} \quad \text{on } \operatorname{Ran} U \oplus Y.$$

Equality $(G - D_W)U = 0$ and self-adjointness give

$$G - D_W = 0 \oplus A_\perp$$

in the same decomposition. Completing the gamma square gives

$$\|(G+\beta)^{1/2}(Ua + v)\|^2 = \|H_\beta^{1/2}(a + H_\beta^{-1} C^* v)\|^2 + \|M_\beta^{1/2} v\|^2.$$

The infimum over a is at least $\beta \|v\|^2$, since $G \geq 0$ and $\|Ua + v\|^2 = \|a\|^2 + \|v\|^2$. Thus $M_\beta \geq \beta$, and the lift in (34) is an isometry into $N_\beta H_{\mathrm{odd}}$. It is onto: for $z \in N_\beta H_{\mathrm{odd}}$, $x = F_\beta z = Ua + v$ lies in the form domain, and $B_\beta^* z = 0$ gives $H_\beta a + C^* v = 0$. Hence $x = J_\beta v$.

Write $x = J_\beta M_\beta^{-1/2} v$. The form of T_β at $\mathcal{O}_\beta v$ is the form of $G - D_W + \beta$ at x. Writing $\mathfrak{a}_\perp$ for the form of $A_\perp$, the exact kernel decomposition gives

$$\langle T_\beta \mathcal{O}_\beta v, \mathcal{O}_\beta v \rangle = \mathfrak{a}_\perp (M_\beta^{-1/2} v) + \beta \|M_\beta^{-1/2} v\|^2 + \beta \|P_0 x\|^2.$$

By (6), the form of $E_{m,\beta}$ at the same vector is exactly $\beta\|U^*F_\beta\mathcal{O}_\beta v\|^2 = \beta\|P_0x\|^2$. Subtracting proves (35).

Since $G_\perp + \beta - M_\beta = CH_\beta^{-1}C^*$ is bounded, their strictly positive form norms are equivalent for fixed β. This makes V_β and its inverse bounded and proves the stated congruence. The block inverse of the strictly positive gamma matrix yields

$$U^*(G+\beta)^{-1}U - H_\beta^{-1} = H_\beta^{-1}C^*M_\beta^{-1}CH_\beta^{-1}.$$

Also

$$V_\beta^*V_\beta = I + M_\beta^{-1/2}CH_\beta^{-1}C^*M_\beta^{-1/2}.$$

The nonzero spectra of the two finite-rank products give the norm identity in (36). For $0 < \beta \le 1$, the eigenvalues of H_β lie between μ_m and $\|H_0\| + 1$. Theorem 1.1 and (9) now give the two-sided order in (36). □

8 Scope and relation to spectral positivity

The positive pivot βH_β^{-1} has norm of order β for a fixed frame. The subtracted correction (6) has the larger order $\beta(\log(1/\beta))^{4m+5/2}$. It nevertheless tends to zero. Thus a bound $C_m\beta$ on this particular correction cannot hold. A sign conclusion for (31) requires control of the retained complementary block at the same scale or an exact cancellation within that block. The correction alone is not a negative direction of the full Weil form. Its magnitude also does not exclude a different normalization or elimination strategy. Proposition 7.1 gives such a strategy explicitly: it removes the finite kernel block first, while the same logarithmic growth appears in the change of gamma metric. Neither normalization determines the sign of $A_\perp$.

Recent work of Kim et al. [6] treats numerical finite-interval realizations of Suzuki's operator and its archimedean spectral law. That setting and the present weighted whole-line resolvent have different parameters and domains. The estimate here concerns $\beta \downarrow 0$ for an exact fixed theta-jet span; it is not a finite-interval eigenvalue computation.

The contribution proved here is the two-sided resolvent and coupling order with a constructive range approximation. This article makes no assertion of priority over all related weighted resolvent estimates. It provides no uniform constants for a growing frame, no matching pointwise spectral-mass lower bound, and no sign for $\mathcal{C}_{m,\beta}$. In particular it proves neither unconditional positivity of all Li coefficients nor RH.

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